

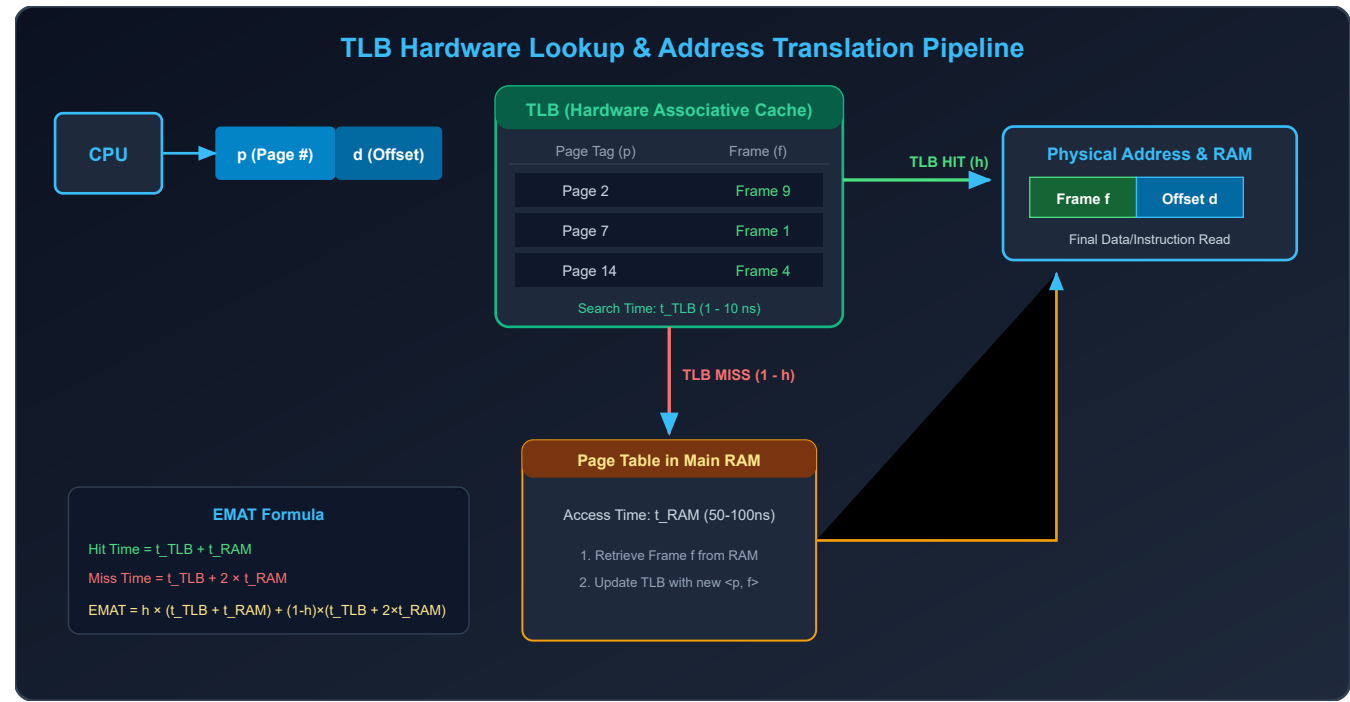
Module 05: Translation Lookaside Buffer (TLB) and EMAT

1. Pure Paging Overhead Problem

Pure paging me har instruction aur data access ke liye do memory references lagte hain: Page table lookup (100ns) + Data access (100ns) = 200ns. Is 100% latency penalty ko door karne ke liye CPU me **TLB (Translation Lookaside Buffer)** hardware cache add hota hai.

2. TLB Hit vs Miss & EMAT Formula

Condition	Probability	Access Formula	Example (t_TLB=20ns, t_RAM=100ns)
TLB Hit	h	t_TLB + t_RAM	20 + 100 = 120 ns
TLB Miss	1 - h	t_TLB + 2 × t_RAM	20 + 200 = 220 ns
EMAT	Total	$h(t_{\text{TLB}} + t_{\text{RAM}}) + (1-h)(t_{\text{TLB}} + 2 \times t_{\text{RAM}})$	$h=0.8 \rightarrow \mathbf{140\ ns} \mid h=0.98 \rightarrow \mathbf{122\ ns}$



3. Protection & Valid-Invalid Bits

- **Valid (1):** Page process ke valid address space me map hai.
- **Invalid (0):** Illegal address access (Hardware trap / Segmentation Fault).
- **Permission bits:** Read (r), Write (w), Execute (x) enforce protection.